

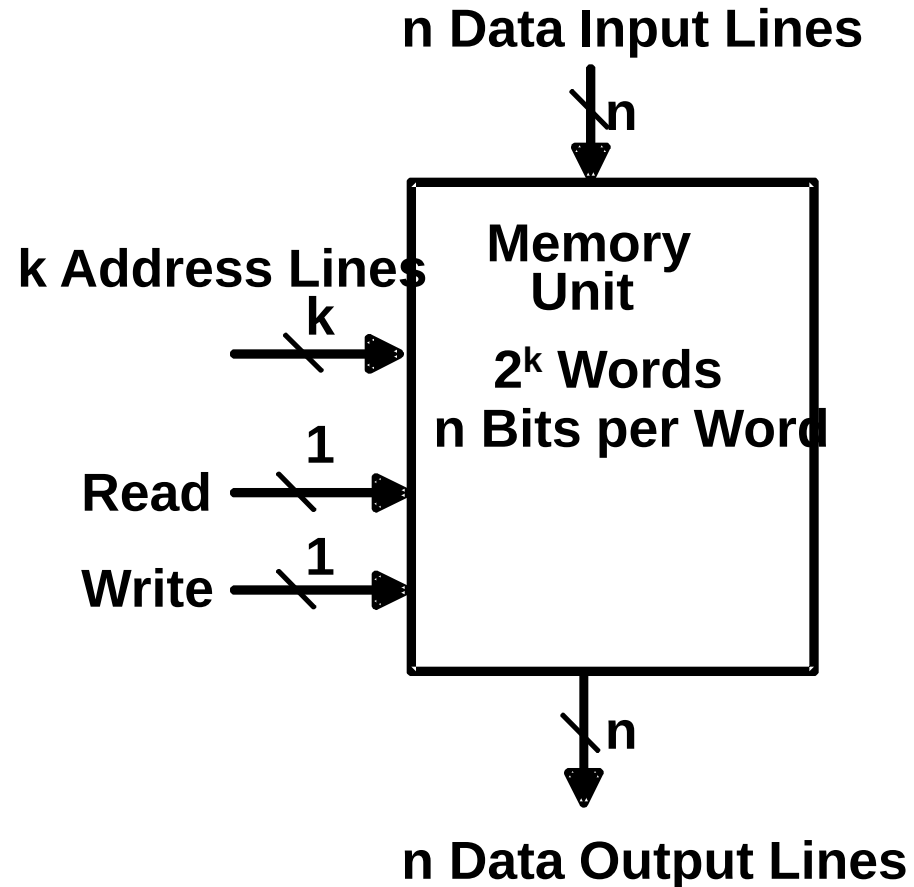
Memory Design

RAM CHIPS

- Features:
 - RAM Chips is suited for the communication with the CPU if it has one or more control inputs that select the chip only when needed
 - Bidirectional Databus – that allows the data either from CPU to memory or memory to CPU
 - Bidirectional Data Bus can be constructed with three state Buffers
 - » Logic 0
 - » Logic 1
 - » High Impedence state

Memory Block Diagram

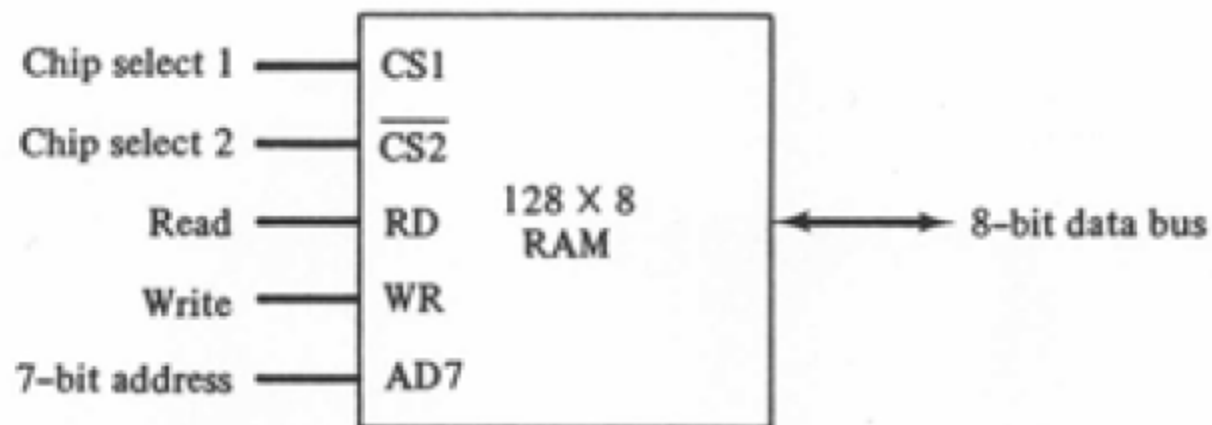
- A basic memory system is shown here:
- k address lines are decoded to address 2^k words of memory.
- Each word is n bits.
- Read and Write are single control lines defining the simplest of memory operations.



RAM Chips

- Capacity of the Memory is 128 words of eight bits
- 7 bit Address and 8 –bit bidirectional data bus
- Read & Write -Memory operation
- Chip Select control Inputs are for enabling the chip only when it is selected by the microprocessor

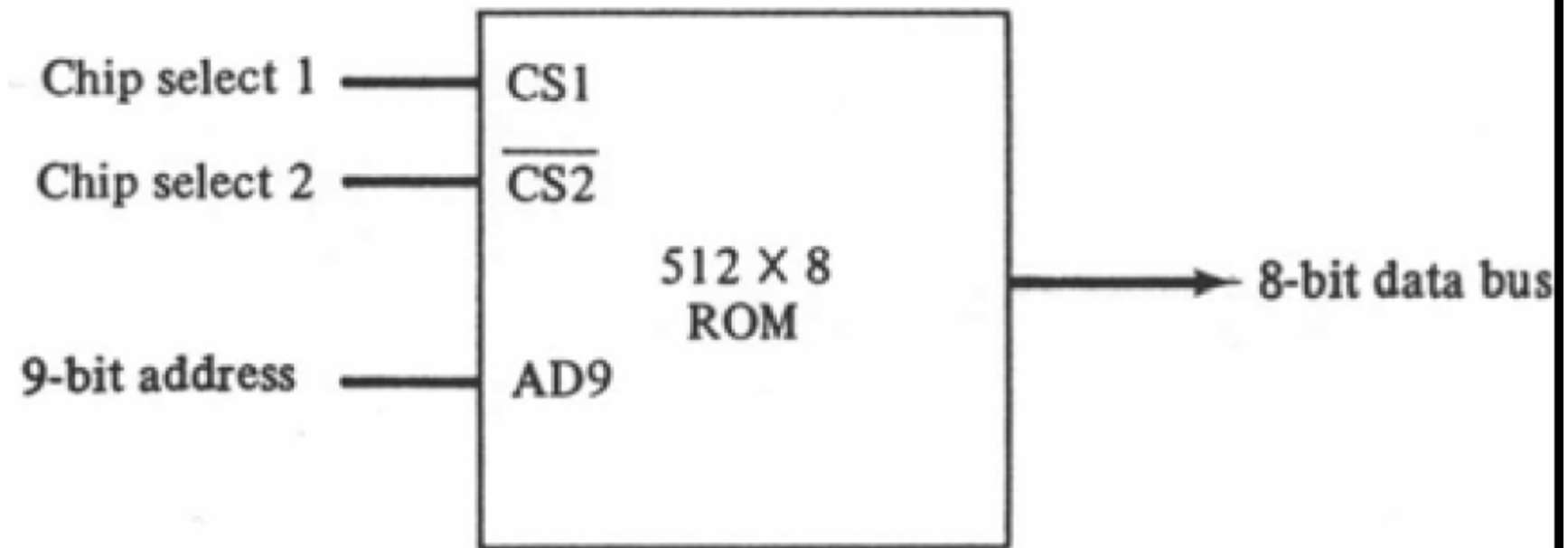
RAM Chip



(a) Block diagram

CS1	$\overline{CS2}$	RD	WR	Memory function	State of data bus
0	0	x	x	Inhibit	High-impedance
0	1	x	x	Inhibit	High-impedance
1	0	0	0	Inhibit	High-impedance
1	0	0	1	Write	Input data to RAM
1	0	1	x	Read	Output data from RAM
1	1	x	x	Inhibit	High-impedance

ROM Chip

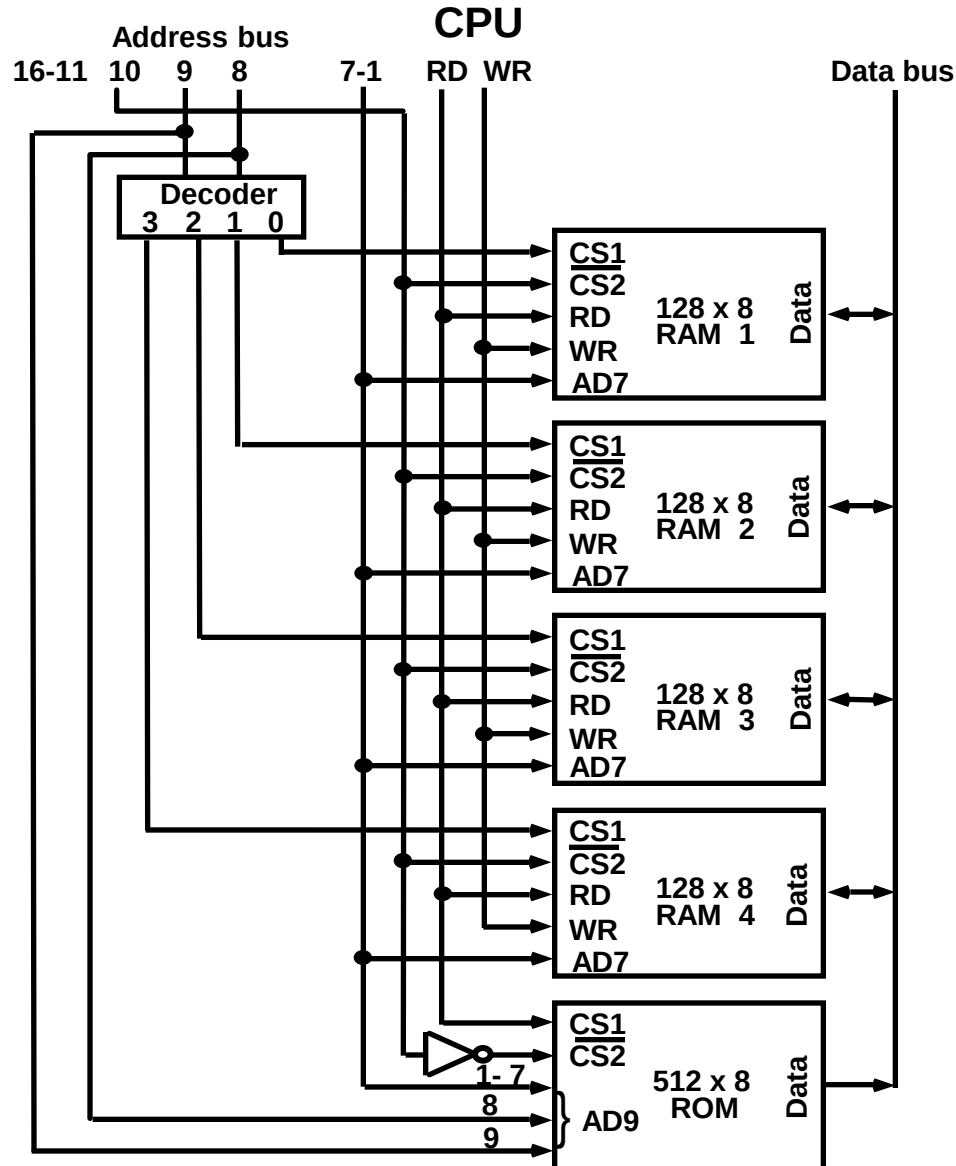


Memory Address Map

Component	Hexadecimal address	Address bus									
		10	9	8	7	6	5	4	3	2	1
RAM 1	0000–007F	0	0	0	x	x	x	x	x	x	x
RAM 2	0080–00FF	0	0	1	x	x	x	x	x	x	x
RAM 3	0100–017F	0	1	0	x	x	x	x	x	x	x
RAM 4	0180–01FF	0	1	1	x	x	x	x	x	x	x
ROM	0200–03FF	1	x	x	x	x	x	x	x	x	x

Address space assignment to each memory chip

Memory Connection to CPU



- RAM and ROM chips are connected to a CPU through the data and address buses
- The low-order lines in the address bus select the byte within the chips and other lines in the address bus select a particular chip through its chip select inputs

Memory Design

- Available Memory chip Size $M_{N, W}$: $N \times W$
- Required memory size: $N' \times W'$, Where $N' \geq N$ and $W' \geq W$
- Required number of $M_{N, W}$ chips: $p \times q$,
Where $p = \lceil N' / N \rceil$ and $q = \lceil W' / W \rceil$

Memory design

There are 3 types of organizations of $N' \times W'$ that can be formed using $N \times W$

- $N' = N$ and $W' > W \Rightarrow$ increasing the word size of the chip
- $N' > N$ and $W' = W \Rightarrow$ increasing the number of words in the memory
- $N' > N$ and $W' > W \Rightarrow$ increasing both the number of words and number of bits in each word.

There are different types of organization of $N^1 \times W^1$ –memory using $N \times W$ –bit chips

How many 1024x 8 RAM chips are needed to provide a memory capacity of 2048 x 8?

Case 1: If $N' > N$ & $W' = W$

Increase number of words by the factor of $p = \left\lceil \frac{N'}{N} \right\rceil$

How many 1024x 4 RAM chips are needed to provide a memory capacity of 1024 x 8?

Case 2:

If $N' = N$ & $W' > W$

Increase the word size of a Memory by a factor of $q = \left\lceil \frac{W'}{W} \right\rceil$

How many 1024x 4 RAM chips are needed to provide a memory capacity of 2048 x 8?

Case 3: If $N' > N$ & $W' > W$

Increase number of words by the factor of p &

Increase the word size of a Memory by a factor of q

Memory design – Increasing the word size

- **Problem - 1**
- Design 128×16 - bit RAM using 128×4 - bit RAM
- Solution: $p = 128 / 128 = 1$; $q = 16 / 4 = 4$
- Therefore, $p \times q = 1 \times 4 = 4$ memory chips of size 128×4 are required to construct 128×16 bit RAM

S.No	Memory Type	$N \times W$	$N^1 \times W^1$	p	q	$p * q$	x	y	z	Total
1	RAM	128×4	128×16	1	4	4	7	0	0	7

x – number of address lines

y ($p = 2^y$) – to select one among the same type of memory

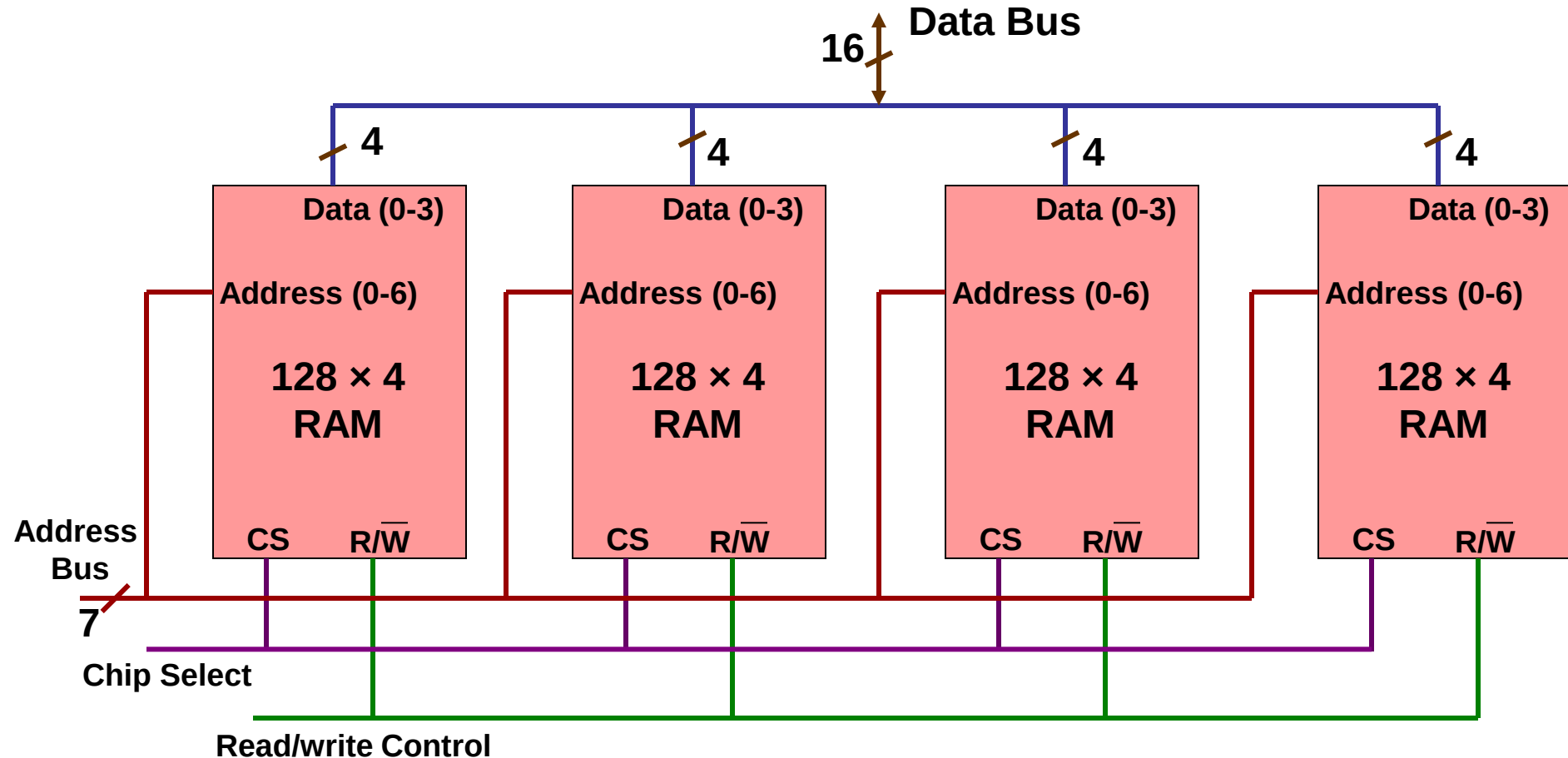
z – to select the type of memory

Memory Address Map

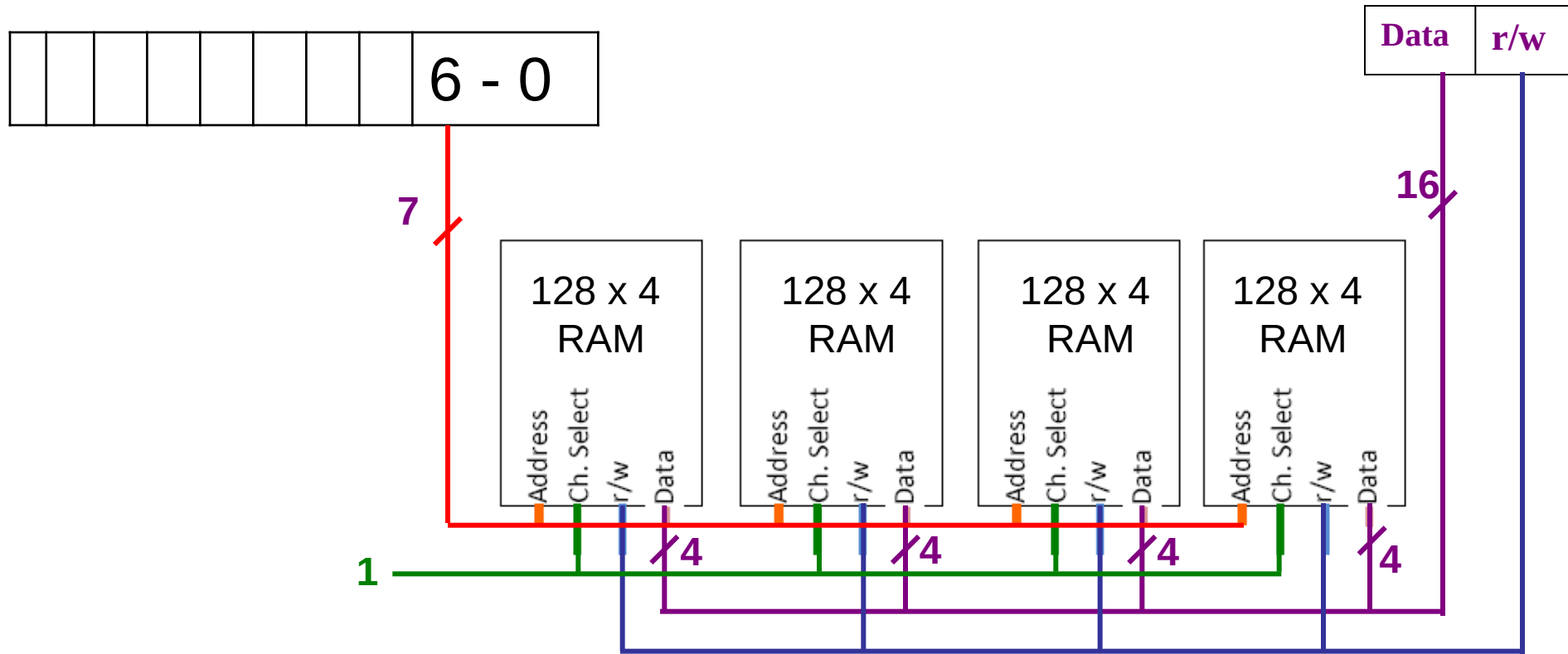
Component	Hexadecimal address		Address Bus															
	From	To	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RAM 1.1	0000	007F										1	1	1	1	1	1	1
RAM 1.2	0000	007F										x	x	x	x	x	x	x
RAM 1.3	0000	007F										x	x	x	x	x	x	x
RAM 1.4	0000	007F										x	x	x	x	x	x	x

Substitute 0 in place of x to get 'From' address and 1 to get 'To' address

Memory design – Increasing the word size



Memory Design



Memory Design – Increasing the number of words

- Problem - 2

- Design 1024×8 - bit RAM using 256×8 - bit RAM
- Solution: $p = 1024 / 256 = 4$; $q = 8 / 8 = 1$
- Therefore, $p \times q = 4 \times 1 = 4$ memory chips of size 256×8 are required to construct 1024×8 bit RAM

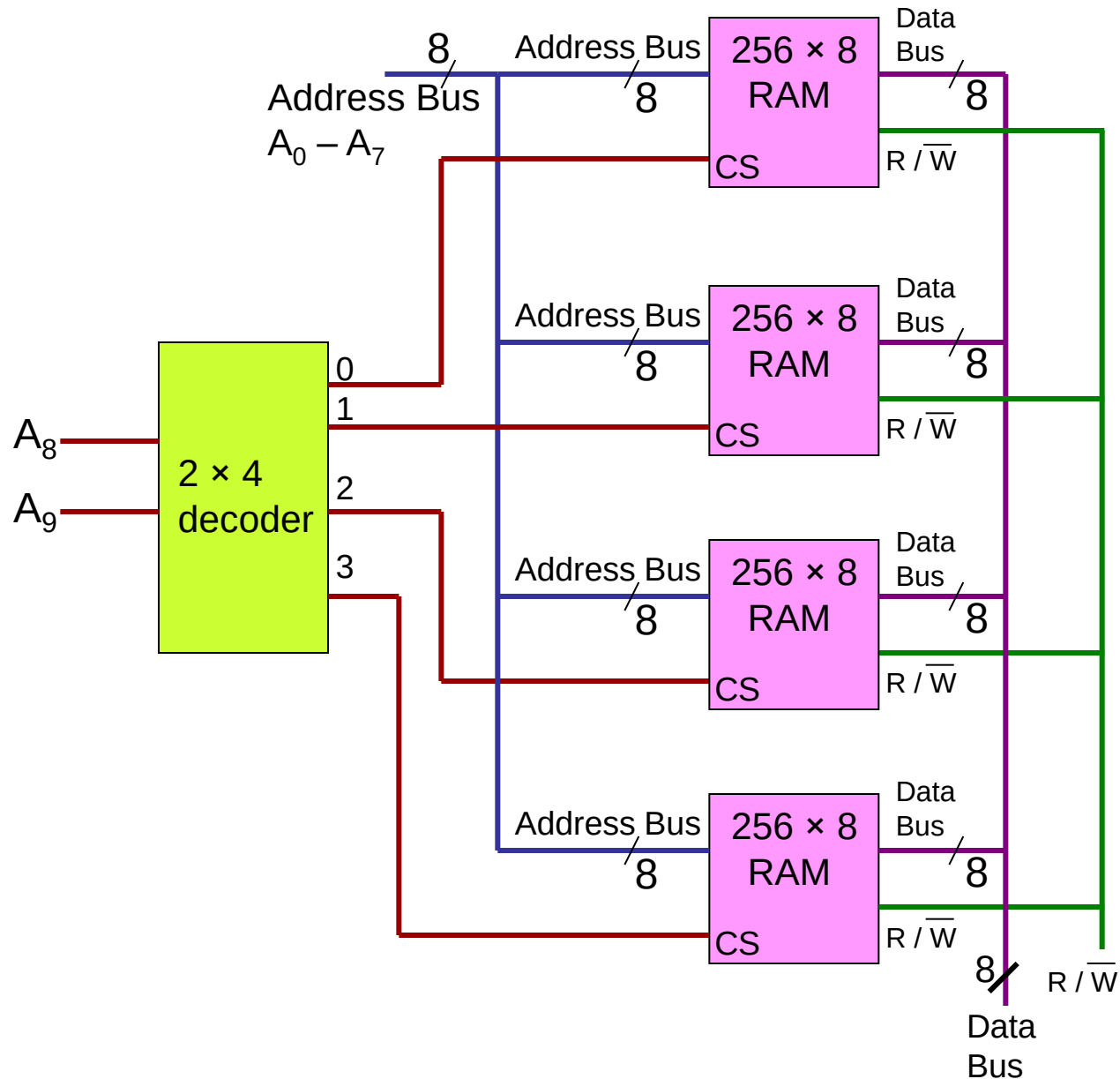
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Memory Address Map

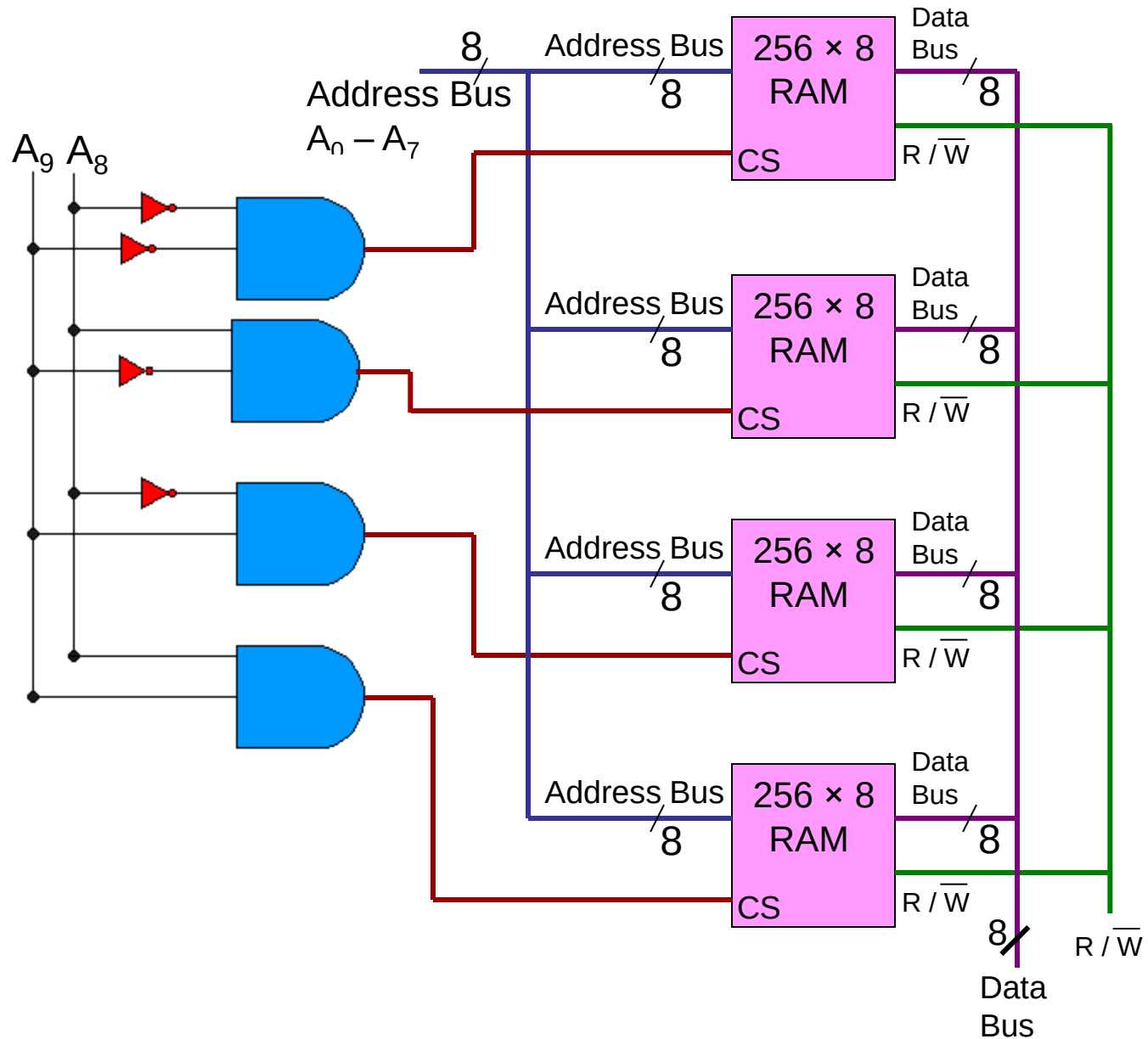
Component	Hexadecimal address		Address Bus																	
	From	To	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0		
RAM 1	0000	00FF							0	0	0	0	0	0	0	0	0	0		
RAM 2	0100	01FF							0	1	x	x	x	x	x	x	x	x		
RAM 3	0200	02FF							1	0	x	x	x	x	x	x	x	x		
RAM 4	0300	03FF							1	1	x	x	x	x	x	x	x	x		

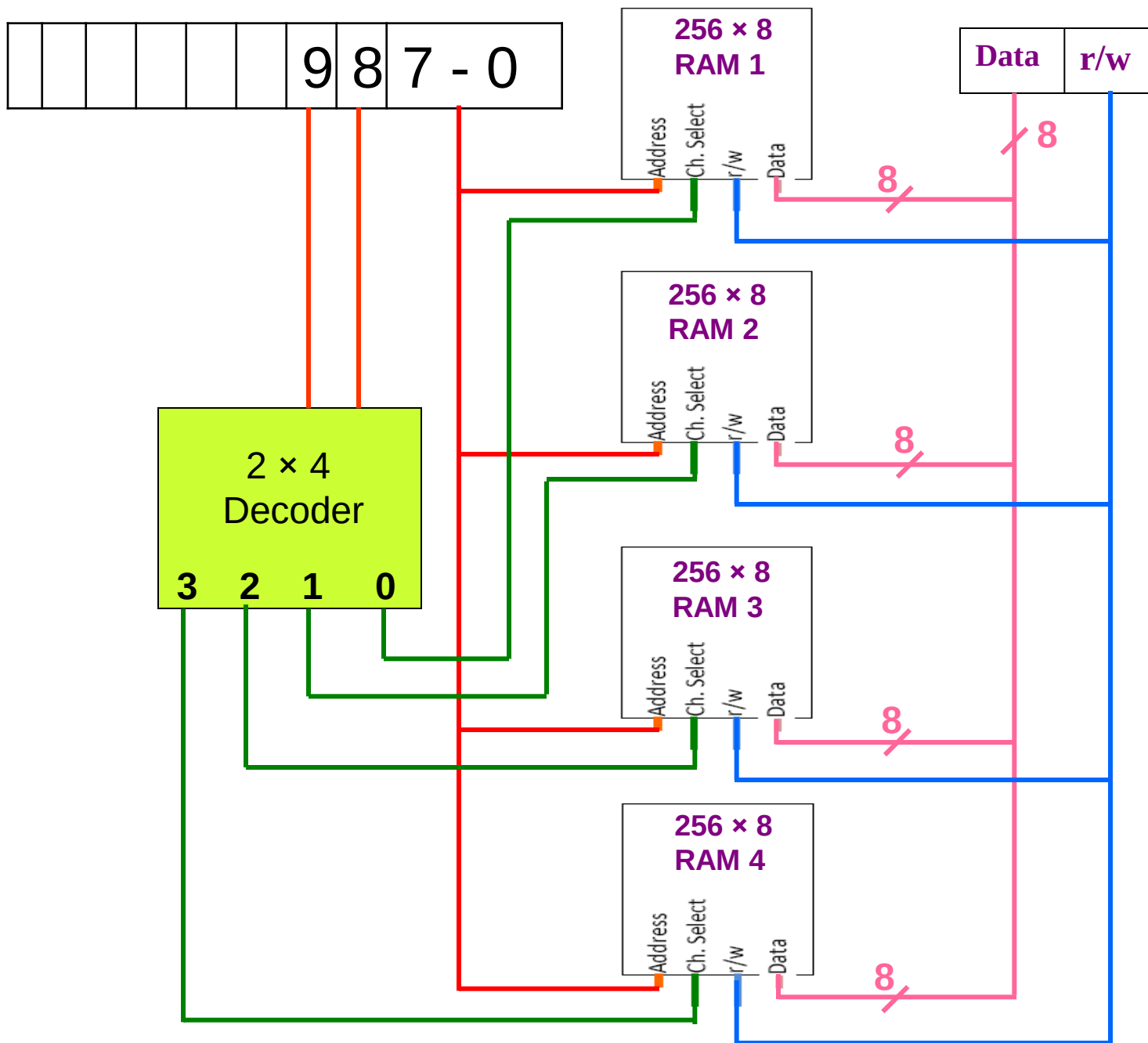
Substitute 0 in place of x to get 'From' address and 1 to get 'To' address

Memory Design – Increasing the number of words



Design with gates





Memory Design

- Problem - 3
- Design 256×16 – bit RAM using 128×8 – bit RAM chips

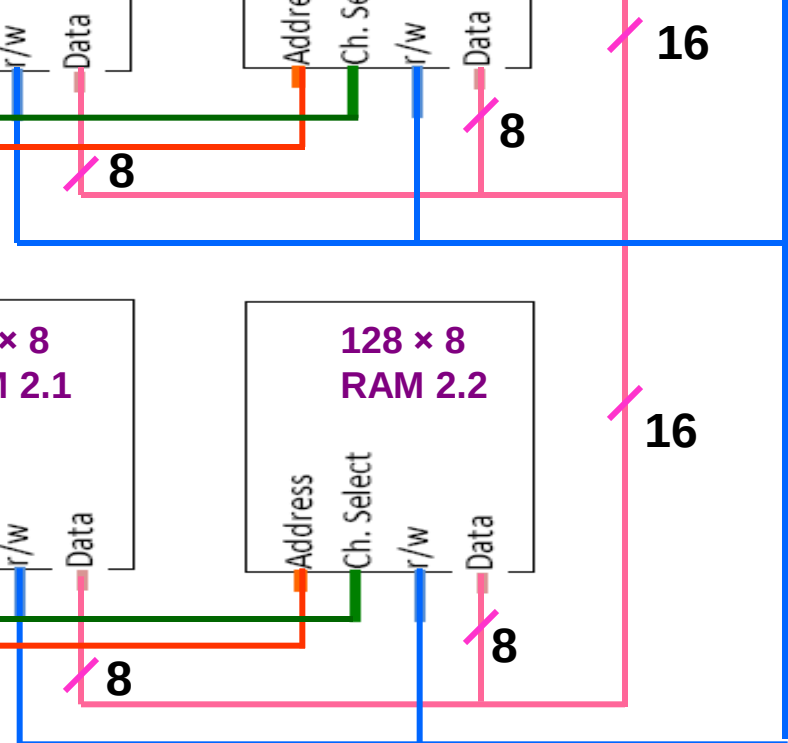
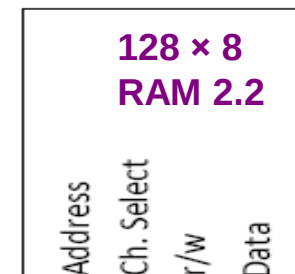
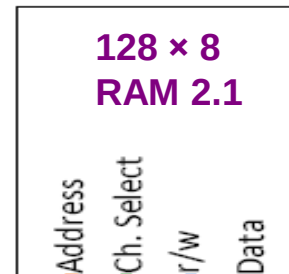
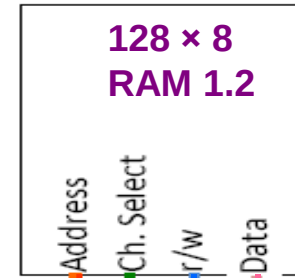
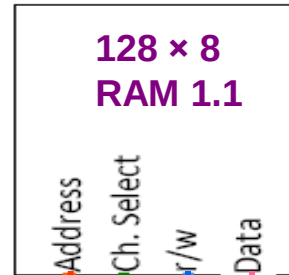
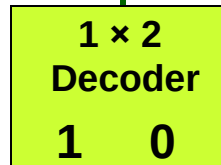
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Memory Address Map

Component	Hexadecimal address		Address Bus																
	From	To	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
RAM 1.1	0000	007F									0	x	x	x	x	x	x	x	
RAM 1.2	0000	007F									0	x	x	x	x	x	x	x	
RAM 2.1	0080	00FF									1	x	x	x	x	x	x	x	
RAM 2.2	0080	00FF									1	x	x	x	x	x	x	x	



Address Bus



Memory Design

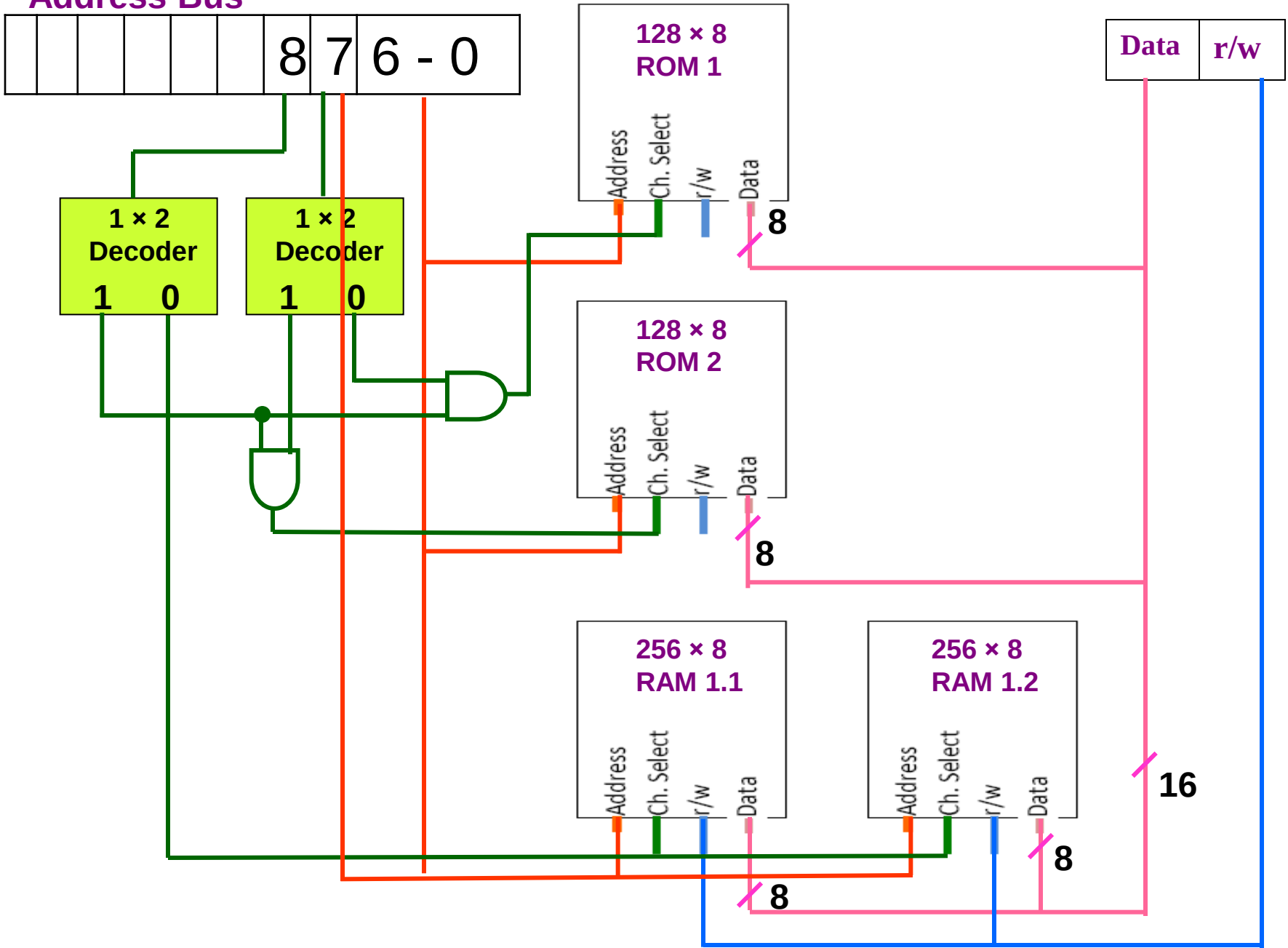
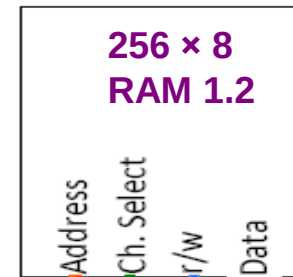
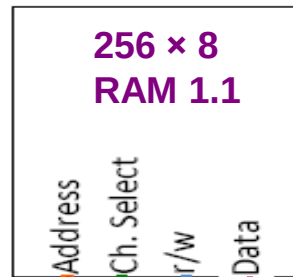
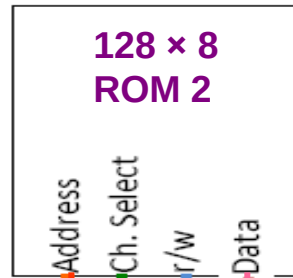
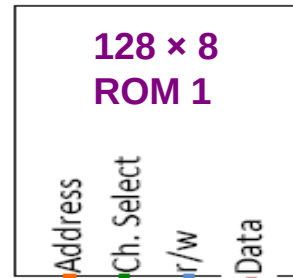
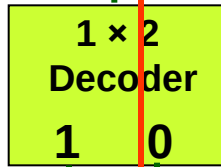
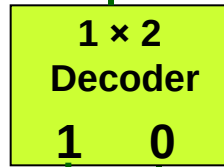
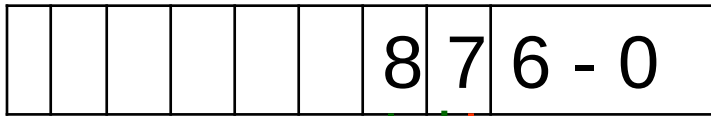
- Problem - 4
- Design 256×16 – bit RAM using 256×8 – bit RAM chips and 256×8 – bit ROM using 128×8 – bit ROM chips.

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Memory Address Map

Component	Hexadecimal address		Address Bus																
	From	To	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
RAM 1.1	0000	00FF								0	x	x	x	x	x	x	x	x	
RAM 1.2	0000	00FF								0	x	x	x	x	x	x	x	x	
ROM 1	0100	017F								1	0	x	x	x	x	x	x	x	
ROM 2	0180	01FF								1	1	x	x	x	x	x	x	x	

Address Bus



Example 1

A computer employs RAM chips of 1024×8 and ROM chips of 2048×4 . The computer system needs $2K \times 16$ of RAM, and $2K \times 16$ ROM and an interface unit with 256 registers each. A memory-mapped I/O configuration is used. The two higher-order bits of the address bus are assigned 00 for RAM, 01 for ROM, and 10 for interface..

- a). How many RAM and ROM chips are needed?
- b). Draw a memory-address map for the system and Give the address range in hexadecimal for RAM, ROM
- c). Develop a chip layout for the above said specifications

References

Text Book(s)

- M. M. Mano, Computer System Architecture, Prentice-Hall, 2004
- J. P. Hayes, Computer system architecture, McGraw Hill, 2000